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To cite this article: I Johnson *et al* 2024 *Environ. Res. Commun.* **6** 105031

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Environmental Research Communications

OPEN ACCESS

RECEIVED
30 May 2024

REVISED
24 September 2024

ACCEPTED FOR PUBLICATION
7 October 2024

PUBLISHED
23 October 2024

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PAPER

Deep-millet: a deep learning model for pearl millet disease identification to envisage precision agriculture

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Keywords: pearl millet, foliar diseases, convolution neural network, optimization-metrics, mobile app, deep learning

Abstract

Plants are integral to human sustenance, serving as fundamental sources of sustenance, materials, and energy, crucial for economic prosperity. However, their productivity and yield are increasingly threatened by pests and diseases, exacerbated by shifting climatic conditions. Pearl millet, a vital crop in Africa and Asia, is particularly susceptible to a range of diseases including downy mildew, rust, ergot, smut, and blast, posing significant risks to crop yield and quality. Timely and accurate disease identification is paramount for effective management strategies. Traditional methods of disease detection relying on visual identification are laborious, costly, and often require specialized expertise, presenting formidable challenges for farmers. In this study, we propose a novel mobile application integrating a robust Deep Learning (DL) model for the automated identification of pearl millet leaf diseases, employing advanced computer vision techniques. A Convolutional Neural Network (CNN) architecture, named Deep Millet, was trained on a comprehensive dataset comprising 3441 field images depicting pearl millet leaves in both healthy and diseased states. It consists of fewer but more effective layers, which are optimized to extract the most pertinent features from the RGB images Comparative analysis against pre-trained models, including AlexNet, ResNet50, InceptionV3, Xception, NasNet mobile, VGG16, and VGG19, was conducted to evaluate the performance of the proposed model. Results demonstrate that Deep Millet achieved superior accuracy, completing training in a mere 240 s and yielding an impressive accuracy rating of 98.86%, surpassing current state-of-the-art models.

1. Introduction

Agriculture is the primary method of food production, upon which the world's economic growth relies. All living organisms depend on this food source; however, plants are increasingly susceptible to numerous diseases due to various climatic changes. Many biotic factors, including fungi, bacteria, viruses, and phytoplasma, contribute to disease occurrences under natural conditions. For instance, fungi can induce leaf spots, blight, powdery mildew, anthracnose, wilt, and root rots in pearl millet. Similarly, bacteria may cause leaf spots, blights, cankers, galls, and wilt diseases, while viruses can lead to symptoms such as mosaic, mottling, necrosis, crinkling, curling, and wilting in susceptible hosts. Additionally, certain abiotic factors such as water stress, nutrient deficiency, and extreme temperatures may mimic the symptoms of biotic factors, making diagnosis challenging without expert support. The cultivation of major crops like rice and millet is profoundly affected by plant diseases, underscoring the critical importance of early detection and timely intervention. Farmers typically seek guidance from agricultural experts to achieve this, but this process can be burdensome, especially in vast farming

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areas where expert field inspections are time-consuming. This challenge is particularly acute in rural regions, where farmers often encounter difficulties accessing expert assistance, sometimes requiring lengthy journeys. Failing to promptly recognize and address plant diseases can exacerbate their severity, resulting in diminished crop yields and potential economic hardships for farmers. Hence, there is an urgent need for innovative solutions that leverage technology to facilitate early disease detection, equipping farmers with the knowledge and tools necessary to effectively protect their crops. The timely detection of diseases is crucial, thus Artificial Intelligence (AI)-driven farming has gained prominence in recent research.

Pearl millet holds significant importance within the millet family due to its ability to thrive in high temperatures, with minimal water, and in low soil fertility conditions. Unlike other crops, pearl millet exhibits exceptional resilience to such extreme environmental factors. Widely cultivated, pearl millet serves as a staple food source for impoverished populations worldwide and is also utilized as fodder for animals. In India, the agricultural sector heavily relies on pearl millet, as the country stands as the foremost producer globally. Recognizing the nutritional value of millets, the Indian Prime Minister has designated them as 'Nutri cereals,' acknowledging their importance as a nutritional treasure[33]. Despite their resilience, pearl millets are susceptible to diseases such as blast, rust, smut, ergot and downy mildew, necessitating timely detection to ensure optimal crop yield.

Therefore, the development of a user-friendly automatic system to aid farmers in early detection of pearl millet plant diseases, without the need for expert intervention, is strongly recommended. CNNs have the capability to autonomously learn features from input data, enabling them to recognize complex patterns and variations in leaf images that may not be discernible to the human eye. This inherent ability makes CNNs a potent tool for detecting leaf diseases that may prove challenging to identify using conventional methods. Various state-of-the-art CNN models, such as AlexNet, VGGNet, ResNet50, Inception, and Xception, are commonly employed for extracting relevant features from plant images. While pre-trained models like these require millions of labeled data to yield optimal results, customized models are preferred for field images due to their ability to cater to specific requirements. Nevertheless, pre-trained models such as AlexNet, ResNet50, VGG, Inception, Xception, and Nasnet serve as benchmarks for evaluating new DL models. These models are frequently utilized as a baseline for assessing the performance of novel models across various computer vision tasks, allowing researchers to gauge the effectiveness of their proposed models and compare their results against established benchmarks in the field.

Deep Learning (DL) algorithms have demonstrated effectiveness when trained on vast datasets. However, the availability of online datasets specific to pearl millet crops is limited. To address this gap, this paper collects field images from various locations in the Coimbatore district. It is noted that the performance of state-of-the-art models on these field images is constrained, resulting in lower accuracy. Therefore, the Deep Millet CNN model is favored over state-of-the-art models, aiming to reduce the false acceptance rate. Furthermore, the model is exported to Teachable Machine, a web-based application capable of capturing real-time images. This enables farmers to view precise disease detection outcomes without requiring prior knowledge of diseases.

The paper makes several significant contributions:

1. The development of a novel system, Deep Millet, based on custom deep transfer learning techniques, enables the identification and categorization of diseases in pearl millet leaves obtained across various locations in the Coimbatore district of Tamil Nadu.
2. The paper provides a thorough comparison of the proposed framework with existing methods for identifying and categorizing plant diseases, offering insights into the comparative performance and efficacy of different approaches.
3. Additionally, the authors implement a web-based Graphical User Interface (GUI) tool, facilitating the creation of custom Machine Learning (ML) categorization models for real-time identification of plant diseases, thereby enhancing accessibility and usability for end-users.

The remaining of the article is presented as follows. Section 2 presents related work published section 3 demonstrates the materials and methodology with dataset preparation and various model used. Section 4 describes the result and discussion. The android application development is shown in section 5. Finally, section 6 concludes the paper.

2. Related works

Identifying downy mildew, rust, blast, ergot, and smut diseases in pearl millet is a difficult task in the field of agriculture. Although previous research has produced insightful findings, there are still a number of gaps and

problems. First, compared to other main staple crops, pearl millet has received relatively less significant investigation, making it the subject of an egregious dearth of studies. The effectiveness of disease detection initiatives is hampered by this information gap, which leads to a lack of specific instruments and methods that are well-suited to the unique properties of pearl millet. One major obstacle is the vast range of disease symptoms that different pearl millet cultivars exhibit in different environmental situations.

Recent advancements in AI methods enable accurate disease and pest attack identification. This paper examines current AI methods for identifying plant leaf diseases. Authors in [1] used ant colony optimization with CNN for citrus fruit leaves with an accuracy of 99.98%. In [2], the author used deep residual CNN with 22 classes of plants and produced an accuracy of 99.58% and was found to be superior to ResNet architecture. The model required more Floating Point Operations (FLOPs) than other state of art models. For the classification of two significant apple leaf diseases, while authors in [3] suggested a Deep CNN with nineteen convolutional layers. The accuracy of test data for identifying apple disease was 99.2%. The deep CNN developed by the authors shows better results for specific crop. In [4], the authors stated that the algorithm used can be implemented into mobile applications for real-time disease detection. This helped farmers to save the crop by taking action against the plant diseases at the appropriate time. It is reported that DL is the best method for this process, which includes CNN and Artificial Neural Network (ANN) models [5–9]. Out of these two, CNN is the best one for image data processing since it contains lots of trainable parameters, and hidden layers and it has lots of predefined models like AlexNet, ResNet, VGGNet, Inception, and others. . These predefined models used in the plant disease detection process were evaluated by classification metrics such as accuracy, specificity, sensitivity, precision, F1score and others.

Many researchers used customized model along with pretrained models for disease prediction and are shown in table 1. Customized CNN model incorporated with segmentation network were implemented for plant disease severity analysis from online images (sui *et al* 2023). Similarly [11] used two publically available datasets namely Plant Village of 9 classes [17], and the Flavia dataset of 32 classes [18]. The datasets were used in compact CNN model 1 and Alex Net model. The performance of the proposed models was improved when they were applied with data augmentation. The accuracies from the Plant Village dataset were 99.45%, and 97.3% for compact CNN models 1 and Alexnet, while, the accuracies from the Flavia dataset were 99.36%, and 99.87% for CNN models 1, and AlexNet, respectively. The result showed that the model performed the same as the AlexNet [17].

Authors in [12] developed an automatic classifier by combining both internets of Things and DL. This system automatically transmitted the images to the cloud; the DL techniques were implemented in the cloud to perform the classification of the received images. The Pearl millet dataset used in this process for training the DL models was prepared by the manual collection of images on the field of two different classes. The authors designed the Custom-Net, which performs equally to the VGG-19 model, but it took lesser size and time for training. Custom-Net would also be very effective if the dataset is small. So, they deployed pre-trained and fine-tuned Custom-Net to the cloud for the classification of the disease.

In their study [14], the researchers introduced a custom CNN model tailored for the plant village dataset, which comprised 12 distinct classes. To enhance accuracy, optimization techniques were applied to the CNN model. They employed image rotation as a strategy to mitigate overfitting. After 20 epochs, the CNN model achieved an impressive accuracy of 93.18%. The authors noted that while the CNN model exhibited excellent accuracy, further increasing the number of epochs could potentially lead to longer training times and an increased risk of model overfitting.

Plant diseases are difficult to detect due to their textures. In this, the authors used Deep Ensemble Neural Networks and other pre-trained models in the automatic system to detect plant diseases. The dataset with 38 different classes was used to train all the models. All the models were fine-tuned by the transfer learning method, and the over-fitting of these models was reduced by applying data augmentation in the dataset [19].

Authors in [20] stated that the farmers should know about the type of plant disease to manage it at an appropriate time. The author developed 10 classes in the plant village dataset for tomato foliar diseases. This dataset was trained by the ML and CNN methods. The accuracy of the K-Nearest Neighbor Algorithm (KNN) in ML was 94.8% and the accuracy of VGG16 in the CNN was 93.5%. The results revealed that the accuracy obtained from ML was better than the neural network methods. The authors stated that the accuracy of CNN could be improved by undergoing various image pre-processing techniques. They also proposed their own lightweight CNN method, which gave the best accuracy of 98.7%. The proposed CNN method performed better than KNN and pre-trained VGG16 [16] used a drone to capture rice crops. However, the system did not take into account real-time prediction.

However, these studies fall short in their ability to detect the factors that are the primary contributors to plant diseases [6, 21–23]. The performance of the current systems can therefore be greatly enhanced, and a new architecture can be provided for automatic disease detection.

In this paper automatic detection is done through

Table 1. Comparative analysis of various CNN models on plant disease detection.

Authors	Objective and dataset	Model	Metrics evaluated
Yousef Methkal Abd Algani 2023 [1]	citrus leaves dataset classification	ACO-CNN optimization approach	Accuracy 99.6% precision 97%, recall 97%, F1 score 97%
Pandian *et al* 2022 [2]	Plant disease classification with 103 classes on *Alovera*, apple, banana, cherry citrus, coffee, corn, egg-plant, etc	ResNet197	accuracy, precision, sensitivity, F1-score and specificity were 99.58%, 99.36%, 99.42%, 99.39%, and 99.27%, respectively
Jwo *et al* 2022 [3]	apple leaf diseases classification	Deep CNN	accuracy 99.2%
Yang *et al* 2020 [10]	Plant leaf classification with Real-time data with 4000 images	VGG16, VGG 19, Inception-ResNetV2	Accuracy of recognition with VGG 19%–92.4%, VGG 16%–91.5%, Inception ResNetV2%–89.6%
Wagle *et al* (2021) [11]	Plant leaf classification with publically available classification - Plant Village of 9 classes and Flavia dataset of 32 classes	CNN models and AlexNet	The accuracies from the Plant Village dataset were 99.45%, and 97.3% for compact CNN models 1 and Alex net, while, the accuracies from the Flavia dataset were 99.36%, and 99.87% for CNN models 1, and AlexNet, respectively
Kundu *et al* (2021) [12]	Real-time field image of pearl millet leaves- rust and blast	Custom net	Achieved Accuracy, precision, recall, and f1 score of 98.78% 99.29%, 98.59%, and 98.64% respectively
Hu, W.-J (2020) [13]	Different crops with 49 disease categories, 10 healthy	Multidimensional feature compensation residual neural network	Achieved 93.96% of accuracy, Recall of 95.4% F1 rating: 96.79%
Khan *et al* 2020 [14]	Tomato Images are taken from both online datasets and real-time images from Tansa Farm	Random Forest (RF) and other models	The accuracy obtained is 99.56%
Solemane Coulibaly *et al* 2019 [15]	Pearl millet downy mildew leaf disease classification	VGG 16	Accuracy 95%, recall of 90.5%, F1 score of 91.75%
Kitpo *et al* 2018 [16]	Rice disease classification with 4 classes with images collected from the rice disease dataset	Support vector machine (SVM)	Only image boundaries with disease alone are detected

- Collecting field images of pearl millet from various locations in the Coimbatore district,
- Creating a dedicated Deep Millet model for disease detection and comparing its performance with state of art models.
- The Deep Millet is further deployed in teachable app to capture field images and to visualize the disease type in real time.

3. Materials and methods

In this section, the author described the dataset preparation, different state-art-of models, the proposed deep millet net, the training procedure, and the classification metrics employed for assessing the model performance.

3.1. Dataset preparation

Constructing datasets is crucial for developing deep learning models, with data annotation playing a central role in the process. In our research, we compiled an image dataset by capturing field trials of crops affected by various diseases. In cases where leaves exhibited multiple diseases, images of distinct sections on the same leaf were obtained. The data collection process utilized a Samsung M30 s mobile phone equipped with a 48-megapixel rear camera featuring a Samsung Bright sensor and 0.8 μm pixel size. Images of both infected and healthy plants were acquired by closely observing leaves ranging from 20 to 60 days old, as disease symptoms typically manifested after the 20th day, and the severity of the disease up to the 60th day significantly impacted yield performance. However, real-world data inherently contains imperfections and often presents challenges related to annotation consistency, such as misclassified class labels and imprecise, duplicated, or missing bounding boxes. To address these challenges, our study involved pathology experts who identified diseased plants based on characteristic leaf shapes, colors, and symptoms, ensuring accurate labeling. Consequently, contemporary

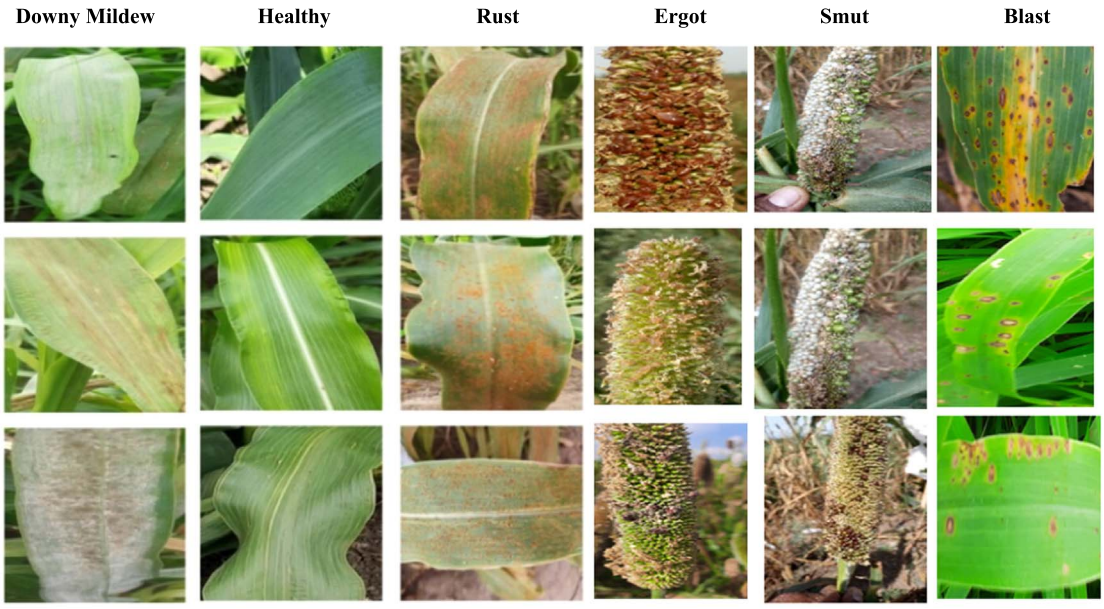

Figure 1. Example of field leaf images in the dataset.

Table 2. Train and test dataset.

Name of classes	Total image	Train dataset	Test dataset
Downy mildew	728	579	149
Rust	960	768	192
Healthy	555	442	113
Blast	450	360	90
ergot	428	343	85
smut	320	256	64
Total	**3441**	**2748**	**693**

machine learning algorithms have integrated various strategies to enhance their robustness in noisy settings. The dataset comprised 728 images of Downy mildew, 960 images of Rust, 555 images of Healthy leaves, 428 images with ergot, 450 images with blast, and 320 images of smut. Given that the captured images varied in size, they were resized to a standard 224 × 224-pixel size. Subsequently, these images were divided into train (80%) and test (20%) datasets. Table 2 illustrates the total number of images in each class, while figure 1 showcases sample images from each class.

3.2. Proposed methodology

The methodology utilizes a dataset comprising 3341 field images of pearl millet gathered from diverse locations in the Coimbatore district. These images are categorized into six distinct classes: downy mildew, rust, blast, ergot, smut, and healthy. The dataset is then partitioned into the common split ratio of 80:20 for training and testing purposes. Initially, all leaf disease images are resized to dimensions of 224 × 224 to ensure uniformity, facilitating compatibility with various pre-trained CNN structures. This preprocessing step aims to expedite the training process when applied to a homogeneous dataset. The classification and validation of pearl millet foliar diseases are conducted using predefined CNN models, including AlexNet, ResNet50, InceptionV3, Xception, NasNet mobile, VGG16, VGG19, and the proposed Deep Millet. The system architecture is illustrated in figure 2.

3.3. Implementation of dl models

Currently, conventional machine learning (ML) approaches are primarily employed for image classification tasks. However, these models have inherent limitations, including restricted capacity, practical application challenges, and susceptibility to overfitting. Traditional feature extraction methods often exhibit low efficiency, resource inefficiency, and incomplete feature representation. As a result, researchers have increasingly turned to Convolutional Neural Networks (CNNs) for recognition studies.

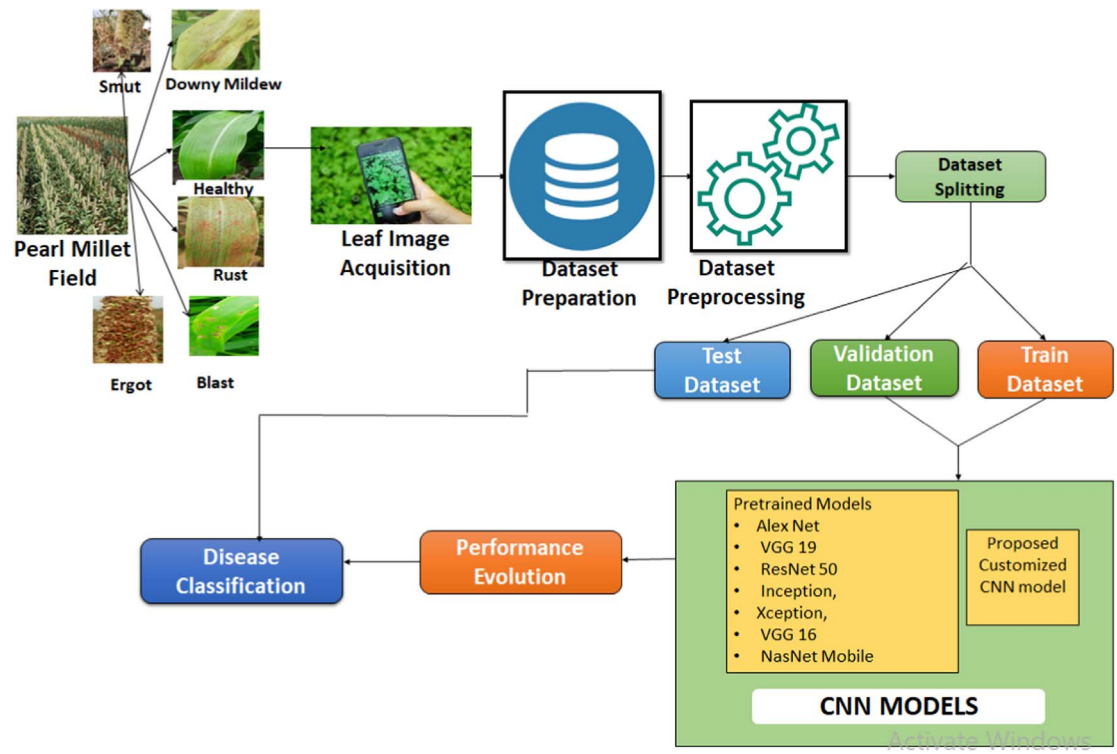

Figure 2. System architecture of the proposed approach.

Figure 3. Pretrained alexnet architecture.

(i) **AlexNet** is a commonly used architecture designed for classification problems. In this model, the input image of the leaf is scaled to $224 \times 224 \times 3$ with zero center normalization, as shown in figure 3. The architecture comprises five convolutional layers, with the primary role of these layers being feature extraction. In each convolutional layer, every neuron is connected to multiple neurons in the neighboring region of the preceding layer. In the AlexNet model, the initial convolutional layer employs 'valid' convolution, while the subsequent four convolutional layers all utilize the same type of convolution (Wu 2021). The specific feature mapping of a single-layer convolution can be described by equation (1):

$$O_i^n = f\left(\sum iO_i^{n-1} \times \ K^n \times B^n\right) \tag{1}$$

where O_i^n is the output of ith neuron after convolution of n layer, K is the convolution kernel and B is bias [24].

(ii) **VGG architecture** : The input image of this 16-layer pre-trained architecture was of size 224 * 244 * 3 and consists of three channels (red, green, and blue). This image is processed, resulting in an output vector of 1000 values given as

$$\mathrm{Y} = \begin{pmatrix} y1 \\ y2\ldots \\ y999 \end{pmatrix} \tag{2}$$

The initial layers of the model consist of 3×3 filters with 64 channels each, utilizing the same padding. Subsequently, a max pooling layer with a stride of (2, 2) is applied. Following the series of convolutional and max-pooling layers, the resulting feature map is shaped as (7, 7, 512). To prepare it for further processing,

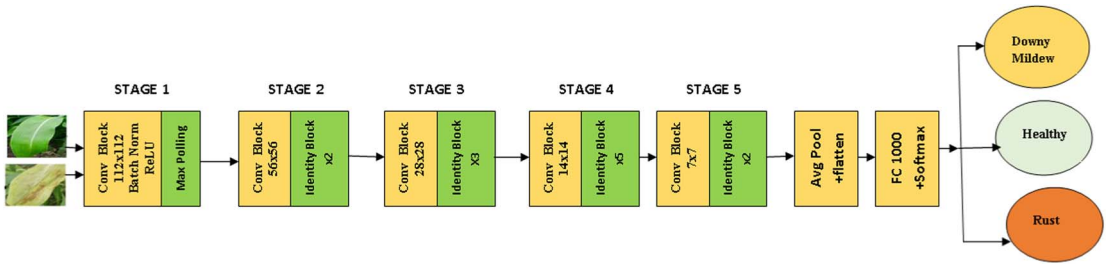

Figure 4. Pretrained ResNet50 architecture.

we flatten this output, transforming it into a feature vector of size (1 × 25088). The flattened feature vector is then passed through three fully connected layers. The initial layer takes the flattened vector as its input, yielding a vector of dimensions (1, 4096) as the output. Subsequently, the second layer similarly generates a (1 × 4096) vector, while the third layer employs the softmax function for the purpose of classifying the input into one of the six distinct classes. Our study leveraged VGG16 and VGG19 architectures for this classification task within the context of these classes.

(iii) ***ResNet 50:*** ResNet 50 architecture consists of five convolution steps and two identities as shown in figure 4. In the first stage, the input image of leaf disease was provided to the first layer, which was made up of 64 separate filtering layers, size two strides, and a 7 * 7 size. The max pooling layer with stride 2 selects the feature region in Stage 2. There are nine layers in the stage 2 convolution layer. Filtering and activation layers were among the 12 and 18 layers, respectively, in stages 3, 4, and 5. 1000 nodes were connected to the average pooling in the end, and one SoftMax function provides the classification result in 6 categories.

(iv) ***INCEPTION V3:*** The output of Inception V3 architecture had the size of 8 * 8 * 2048. The model consists of 42 layers with a size of 3 * 3 and several convolutions. It had 3 modules. Module 1 had a 3 * inception and an input size of 35 * 35 * 288. Module 2 had a 5 * inception and a 17 * 17 * 768 input size. In module 3, the input size was 8 * 8 * 1280 with 2 * inception. In the Inception-v3 model, a batch normalization (BN) layer is added as a regularizer between the auxiliary classifier and the fully connected (FC) layer. Incorporating the BN (Batch Normalization) layer into the model facilitates the use of batch gradient descent, leading to faster training and enhancing the convergence of the deep neural network, as outlined in the equation.

$$Y_i = \alpha X_i + \beta = BN(X_i) \tag{3}$$

with learning parameter α, β

$$X_i = \frac{Xi - \mu}{sqrt(sq(SD) + \varepsilon)} \tag{4}$$

Where μ denotes the average value along a specific dimension, SD represents the standard deviation in each dimension of the feature map, and ε is a constant parameter. Xi is the input vector and Yi is the output. In this paper, The softmax classifier was used to detect the disease.

(v) ***Xception:*** Xception also called extreme inception, is based on the Inception architecture, but instead of using traditional convolutions, it uses depth wise separable convolutions. In traditional convolutions, the same set of filters is applied to each channel of the input tensor, which can be computationally expensive. Depthwise separable convolutions, on the other hand, apply a separate set of filters to each channel of the input tensor, which reduces the number of parameters and makes the model more efficient [13] Let the image size be M × M then kernel size will be mxnxF where F is the filters required for the given output. The total number of multiplication is given by F × N2 × M2, where N2 is the convolution multiplication for single operation [31, 32].

(vi) ***NasNet mobile:*** NasNet Mobile is a lightweight version of the original NasNet architecture, which was designed for larger-scale image recognition tasks. The architecture is based on the idea of 'neural architecture search,' which involves using a reinforcement learning algorithm to automatically search for optimal network architectures. NasNet Mobile consists of a sequence of convolutional, pooling, and fully connected layers, with a total of about 4.9 million parameters. It uses a combination of depthwise separable convolutions, which reduce the number of parameters and computations required, and residual connections, which facilitate gradient flow and improve training stability [11, 13].

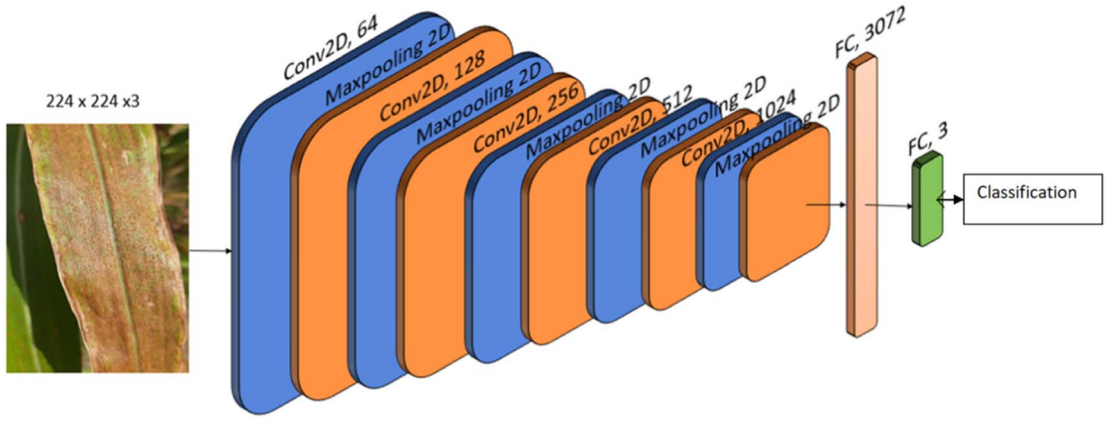

Figure 5. Proposed deep millet architecture.

(vii) ***Deep millet model:*** The models can be finely tuned to capture domain-specific features and patterns, enhancing their performance in specialized tasks. By optimizing the model's architecture, researchers and developers can achieve superior results, faster inference times, and reduced computational overhead. Custom models are particularly valuable in scenarios with limited labeled data. The customized CNN model was designed to predict six classes: Downy mildew, blast, smut, ergot Rust, and Healthy leaf images. The model (figure 5) consisted of five Convolution2D layers with ReLU activation function and five Maxpooling2D layers in sequential order of three sets after that batch normalization was performed and the flattening method was also applied. After flattening a fully connected (dense) layer followed by a dropout and classification layer with SoftMax activation function to classify six classes.

The Convolution layer contains a series of filters that convolute the complete image. The numerous features that capture discriminating patterns to determine the type of plant leaf were learned by each convolution layer in this design. The feature shrinkage was reduced by padding. Reducing feature shrinkage is one of the benefits of padding. Feature shrinkage occurs when the size of feature maps in a neural network is reduced during processing, typically by pooling or convolutional layers. This can lead to loss of information and reduced performance in the neural network. By padding the input data, the feature maps can be preserved or even increased in size, which helps to reduce feature shrinkage and maintain performance. Deep Neural Networks observed distinct feature information from the preceding layer after each gradient update on a batch of data [11]. Furthermore, this input features maps data distribution fluctuates substantially, since the preceding parameter layers were modified throughout the training process. This had an enormous influence on training speed and involved the use of heuristics to figure out how to set up the parameters [25]. ReLU, is a type of activation function that was extensively employed in building neural networks and increases the nonlinearity of CNNs. It was a function that gave output only for positive values given by

$$F(x) = \max(0, x) \tag{5}$$

and for negative values, it returns 0. ReLU was triggered occasionally, simulating the inactivity of actual neurons in response to specific impulses [7, 26]. All the CNN layers had ReLU as the activation function. Only a fraction of the feature map neurons were activated by the Max-Pooling layer. The width and height of the feature map were successfully lowered, while the number of channels was well maintained. The Fully Connected layers were utilized for classification tasks since all the neurons in them were linked to all the neurons in the following layers [15]. The SoftMax function was utilized as the activation function in the output layer of CNN models that predict a multinomial probability distribution.

$$X_i = \frac{e^{X_i}}{\sum X_j} \tag{6}$$

where $j = 1$ to K; $K = 3$ classes and X_i is the input vector of real number

The proposed model employed a consistent filter size of (3×3) across all five CNN layers, a choice supported by prior research highlighting its effectiveness in image classification [27]. These CNN layers were configured with filter sizes of 64, 128, 256, 512, and 1024 to progressively extract hierarchical features from the input data. Max-pooling layers, using a '2-by-2' window with a stride factor of '2', were strategically positioned between the CNN layers for dimensionality reduction. Batch normalization was applied following each Max-pooling layer to enhance training stability. Following the feature extraction process facilitated by the CNN

layers, the classification phase ensued through two Fully Connected (FC) layers. The 1st FC layer featured 4096 neurons with ReLU activation, while the second FC layer consisted of 3 neurons and employed a SoftMax activation function for output extraction.

3.4. Classification metrics

Classification metrics such as True Positive (TP), True Negative (TN), False Positive (FP) and False Negative (FN) are used in the confusion matrix to perform the comparison of different methods. A sample multiclass confusion matrix for Downy mildew. All the rows in the confusion matrix represent actual classes, and all the columns in the confusion matrix represent predicted classes. It showed the number of correctly and incorrectly classified labels [28].

The authors defined the performance matrices based on the classes supplied in the confusion matrix.

Average accuracy: The accuracy metric measures the correct prediction ratio [29]. The accuracy was calculated using equation (7).

$$Accuracy = \frac{True\ Positive + True\ Negative}{True\ Positive + False\ Positive + True\ Negative + False\ Negative} \tag{7}$$

Precision: To compute the percentage of correctly predicted positive patterns in a positive class out of all predicted patterns [29]. The precision was given in equation (8).

$$Precision = \frac{True\ Postive}{True\ Positive + False\ Positive} \tag{8}$$

Recall: Deep learning models, while often powerful, can produce false negatives, potentially leading to missed opportunities or critical errors. By prioritizing recall, deep learning practitioners ensure that their models excel in capturing true positives, minimizing the likelihood of false negatives, and enhancing the model's reliability and utility in critical real-world scenarios. The recall measures the percentage of properly identified positive patterns [29]. The Recall was calculated using equation (9).

$$Recall = \frac{True\ Positive}{True\ Postivie + True\ Negative} \tag{9}$$

F1-Score: The harmonic average of the recall and accuracy values [30]. The formula used to calculate F1-Score is mentioned below (10).

$$F1\ Score = \frac{2*Precision*Recall}{Precision + Recall} \tag{10}$$

4. Results and discussions

The results obtained by the proposed Deep Millet model and seven different pre-defined models: VGG-19, Xception, VGG-16, AlexNet, NasNet-Mobile, Inception-V3, and ResNet-50 with the real-time dataset (3441 images) of 6 classes: Downy mildew, Rust and Healthy were compared and discussed in this section

4.1. Validation and comparison of deep millet model and pre-defined CNN

The performance of the Deep Millet model and other pre-defined CNNs was validated and compared using various classification metrics. The experiments were conducted for 150 epochs for all models to achieve better accuracy. Adam, Stochastic Gradient Descent (SGD), and Root Mean Squared Propagation (RMSprop) optimizers were employed for training and testing the dataset. SGD with momentum (SGDM) incorporates momentum into stochastic gradient descent to improve its performance. RMSprop is an optimizer that dynamically adjusts the learning rate using root mean square propagation. Adam, an optimizer derived from adaptive moment estimation, combines the advantages of momentum and RMSprop for efficient neural network optimization. Upon analyzing the results of the three optimizers, Adam outperformed the others with the highest accuracy of 98.46%, as depicted in table 3. Figures 6(a), 7(a), and 8(a) illustrate the accuracy graph for the Deep Millet model using Adam, SGD, and RMSprop optimizers, respectively. Figures 6(b), 7(b), and 8(b) depict the accuracy graph for the Deep Millet model using Adam, SGD, and RMSprop optimizers, respectively.

The performance of the customized model and all the other six pre-defined CNN classification models were compared with the values of Accuracy, Precision, Recall, and F1-Score and are shown in table 4.

Figure 9 shows the performance metrics reported by deep millet model which performed better with an accuracy value of 98.86% followed by VGG-16 with 97.80%, ResNet-50 with 97.58%, Xception with 96.7%, NasNet-Mobile with 96.7%, Inception-V3 with 93.39%, VGG-19 with 88.77% and AlexNet with 87.22%, respectively.

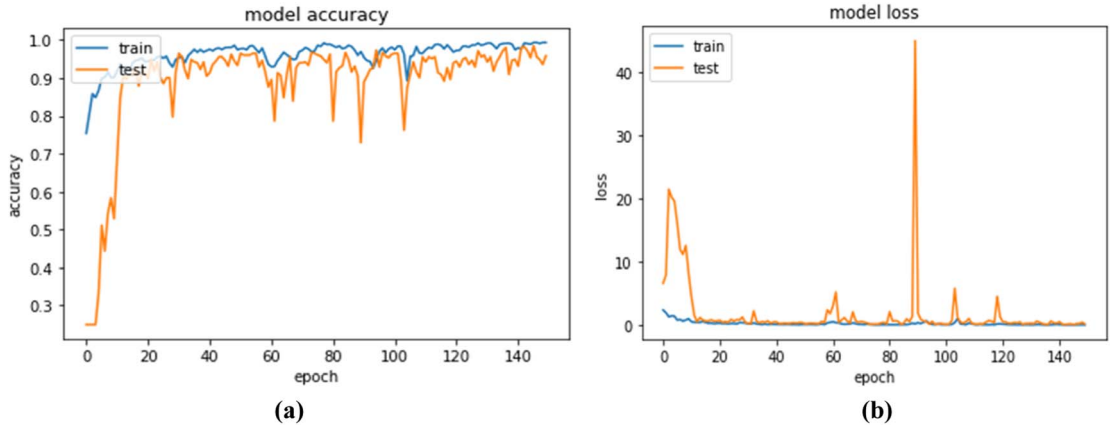

Figure 6. (a) Accuracy graph of deep millet using adam optimizer, (b) accuracy graph of deep millet using adam optimizer.

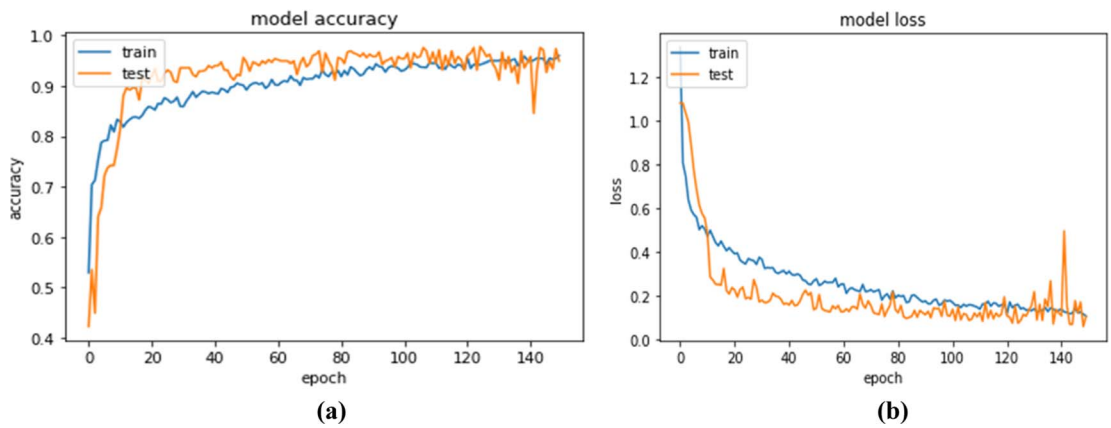

Figure 7. (a) Accuracy graph of deep millet using SGD optimizer, (b) Loss graph of deep millet using SGD optimizer.

Table 3. Accuracy of deep millet with different optimizers on test data.

	Adam	SGD	RMSProp
Accuracy	98.86	96.4	94.12
Model loss	1. 14	4.6	6.98

The time required to train an AI algorithm varies depending on several factors, including the algorithm's complexity, dataset size and quality, available computational resources, and specific training techniques utilized. It's essential to recognize that not all AI algorithms can be trained quickly, and many advanced models require significant time investments, often spanning hours, days, or even weeks to achieve optimal training results. In our research, we found that smaller and less complex models like AlexNet and VGG-16 tend to have shorter training times compared to larger models such as NasNet Mobile, ResNet-50, and Inception V3. Medium-sized models like VGG-19, VGG-16, and Xception may require slightly longer training times than smaller models. To maintain consistency, all models in our study were trained for 20 epochs.

Specifically, for the given dataset, the training times were as follows: AlexNet and VGG-19 took 300 s and 360 s, respectively. Inception V3, NasNet, and Xception required approximately 480 s, 360 s, and 480 s, respectively. Notably, VGG-16 only took 300 s for 20 epochs. In contrast, the proposed customized model demonstrated a shorter training time of 240 s, which is notably less than other models.

To validate the effectiveness of our suggested model in comparison to cutting-edge DL models, including ResNet-50, VGG-16, Xception, NasNet, Inception-V3, VGG-19, and AlexNet, our results indicate that the proposed model achieved a considerably reduced training time of 80 s for 20 epochs, as illustrated in figure 10.

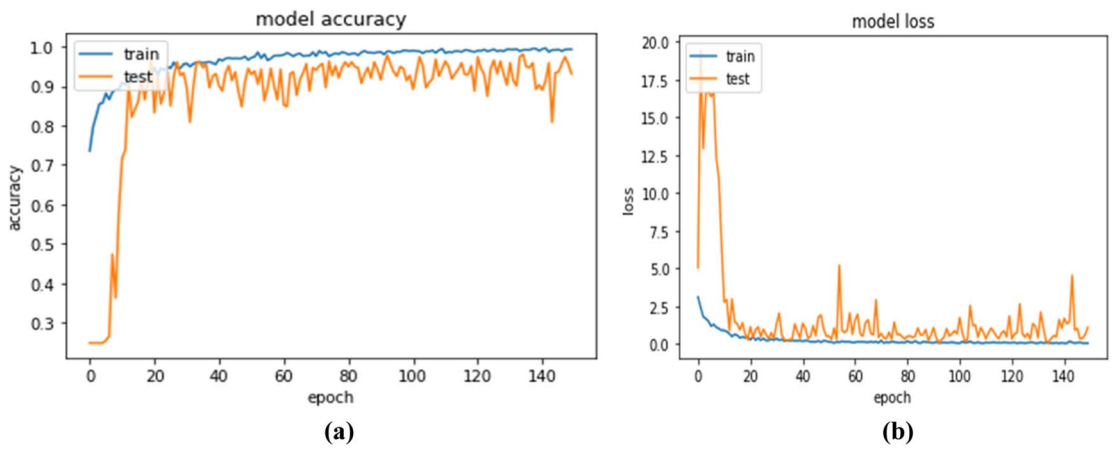

Figure 8. (a) Accuracy graph of deep millet using RMSProp optimizer, (b) accuracy graph of deep millet using RMSProp optimizer.

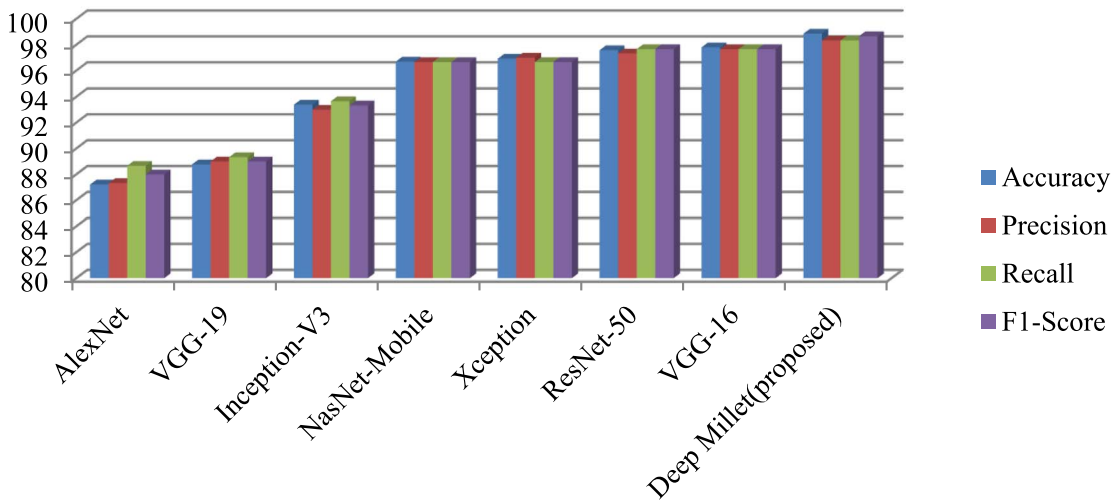

Figure 9. Performance evaluation with state-of-the-art algorithms.

Table 4. Comparative analysis of various DL models.

	Accuracy	Precision	Recall	F1-Score
AlexNet	87.22	87.33	88.66	88.00
VGG-19	88.77	89.00	89.33	89.00
Inception-V3	93.39	93.00	93.66	93.33
NasNet-Mobile	96.7	96.66	96.66	96.66
Xception	96.92	97.0	96.66	96.66
ResNet-50	97.58	97.33	97.66	97.66
VGG-16	97.80	97.66	97.66	97.66
Deep Millet (proposed)	**98.86**	**98.33**	**98.33**	**98.66**

4.2. Comparative analysis with other state-of-the-art schemes

The experimental findings achieved by using the proposed and state of art models were presented by the authors along with the inferences drawn from them. The average accuracy, precision, recall, F1 score, and training time of the proposed model were comparable to those of other DL models. Additionally, the proposed model needs at least 12 s per epoch as training time. It just takes 240 s to finish its training via 20 epochs. Its capacity to achieve classification accuracy equivalent to cutting-edge models with less training time attests to its suitability for use in

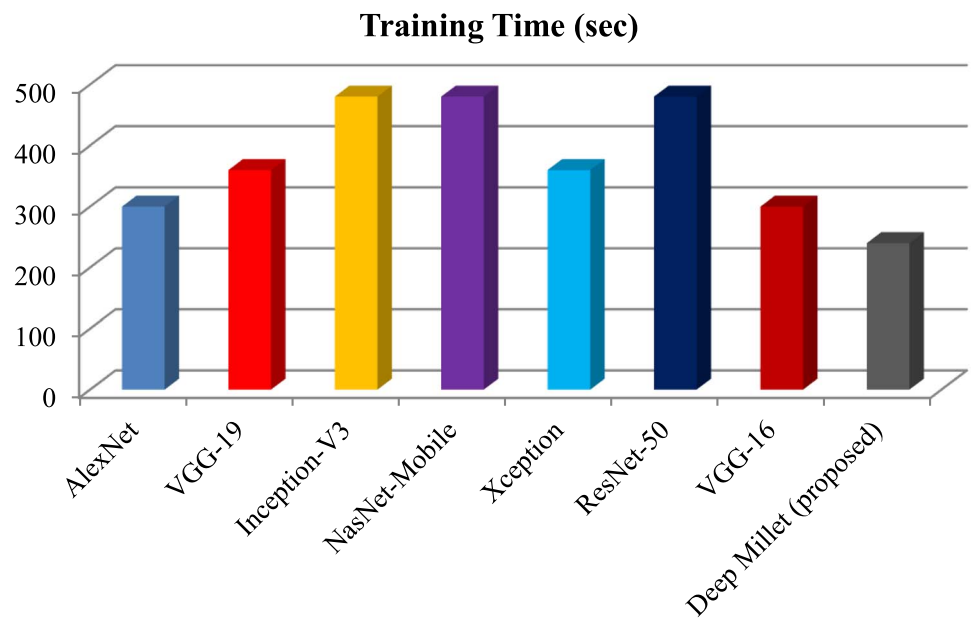

Figure 10. Training time for different Dl algorithms.

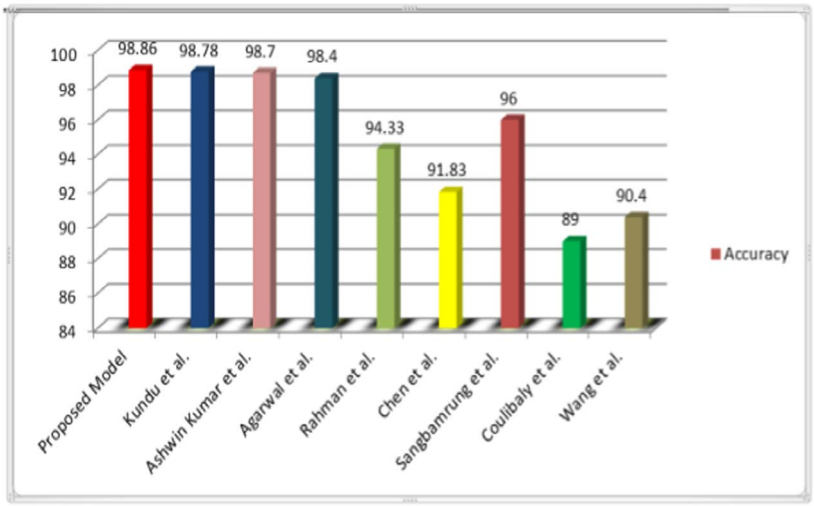

Figure 11. Proposed versus state-of-the-art comparisons.

real-time systems. Furthermore, classifying the rust, smut, ergot and downy mildew-affected samples in real-time was useful for making speedy decisions.

Due to the lesser number of layers, the time required for training and storage for this model is less than other pre-defined models. Many pre-processing techniques on images were undergone to get better results by running the model for fewer epochs.

In the proposed work, the dataset has 6 classes and the Proposed CNN model performs better than other pre- defined models with this dataset by an accuracy value of 98.58%. The proposed model has a lesser number of layers and requires less time for training. The comparative analysis of proposed method with existing method is depicted in figure 11.

5. Android application for deep-millet

In this study, GUI tool was introduced to capture the diseased leaf in real time and to check whether the model predicts the disease accurately [31, 32]. Teachable Machine (TM) is a web-based GUI tool for building unique

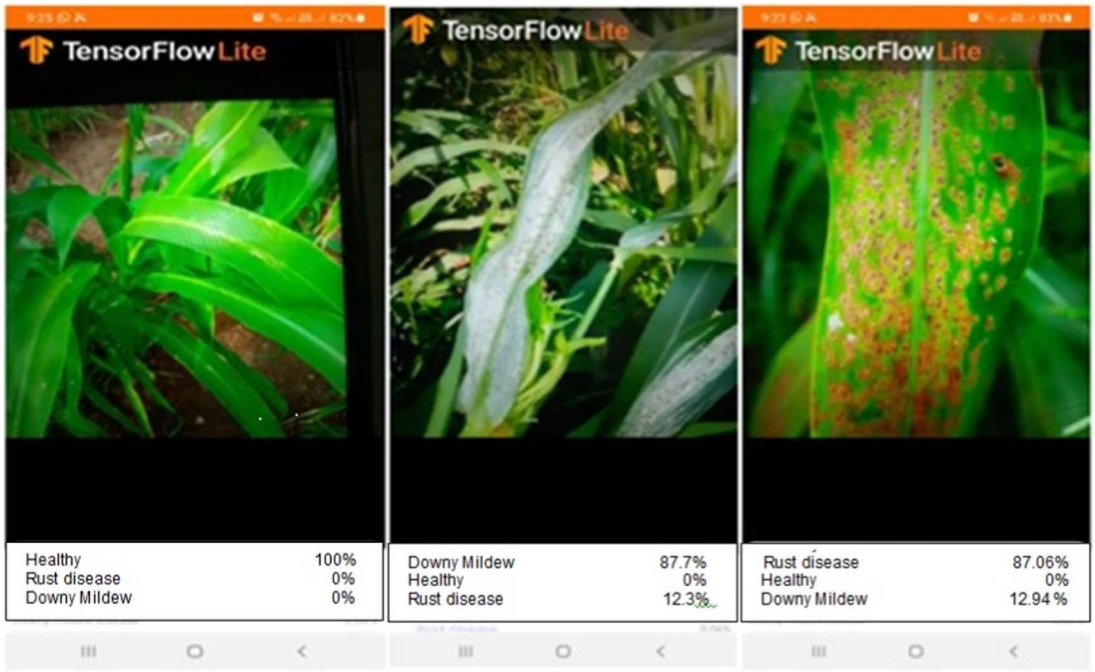

Figure 12. Real time prediction.

ML categorization models. It uses Tensorflow.js, an open-source JavaScript library, to create and train ML models. TM makes use of the transfer learning technique, allowing users to build their ML models on top of the Google MobileNet architecture using their datasets. This study used the Teachable Machine to classify and identify diseases. After successful training and testing of proposed model, the model was converted into .h5 format and then to .tflite file with appropriate tools and libraries. Once the conversion process was completed, the resulting .tflite file was saved in mobile phone. Additionally, a text file was created to store the dataset labels that were used for classification during the training process. This text file contains the necessary information for labeling and categorizing objects or data points. Finally Teachable Machine application was opened on the mobile phone. Within the application, .tflite model file was selected and dataset labels from the text file were imported. The real time detection were initiated using mobile phone's camera directly within the Teachable Machine application. The model's accuracy was then assessed to guarantee the stability of the method. The trained Proposed CNN model was converted into a TensorFlow Lite model. Import the TensorFlow Lite model to the mobile phone, open the app and select the tflite and the labels (6 classes with names) file. Click start, the mobile camera will be opened and show the camera to the plant leaf as this app will show the percentage of accuracy of 3 classes as shown in figure 12.

It is observed that using teachable machine, downy mildew and healthy leaves were detected perfectly with an accuracy of 100% in real time. However, the rust disease has shown an accuracy of 87.06%. The accuracy is reduced due to its coincidence with blast disease and it will be rectified in the future.

Scientific contribution of this work

1. Pearl millet is a rainfed, nutrient rich grain crop though cultivated through out the world there is no disease detection app available so far in this context this is highly useful to the farming community
2. The proposed CNN model demonstrates an accuracy of 98.58% on a dataset. The proposed model has fewer layers. By reducing the number of layers, the model not only becomes more efficient but also requires fewer computational resources, making it particularly well-suited for deployment in environments with limited resources
3. The study emphasizes the implementation of several image pre-processing techniques, which play a crucial role in enabling the model to reach high accuracy with fewer training epochs.
4. The model reduces training time, taking 240 s to complete 20 epochs. This efficiency is crucial for real-time systems, where fast training and inference are vital. This supports rapid decision-making, which is essential for effective crop management.
5. The study highlights the practicality of the proposed model for real-time applications, especially in situations where labeled data and computational resources are limited.

6. Conclusion

In this study, we have introduced a robust DL framework for the automated identification of plant leaf diseases in Pearl Millet crops. Leveraging a dataset comprising 6 classes of Pearl Millet diseases captured via mobile camera, our system utilized computer vision techniques to construct the Deep Millet model, featuring a reduced number of hidden layers for disease classification. Remarkably, our model demonstrated superior accuracy and significantly reduced training time compared to state-of-the-art models. Trained on 3441 images of both diseased and healthy pearl millet leaves, the Deep Millet model achieved an impressive accuracy of 98.66% within a training time of just 240 s. The ability of early detection in case of diseases which spreads in an exponential pattern (leaf spots, blights, rust, powdery mildew, downy mildew etc,) would be highly beneficial for timely intervention and more effective disease management strategies besides reducing the yield losses in crop cultivation. Regarding the early detection of diseases in pearl millet, our analysis indeed supports the possibility of identifying diseases like downy mildew, smut, and ergot at early stages, as the relevant datasets for these diseases were included in our study. However, early detection poses more challenges in the case of rust and blast diseases. These diseases typically manifest initially as reddish-brown spots, making them harder to detect before the visible symptoms become apparent. Detection in these cases relies heavily on the characteristic symptom expression, which might occur later in the disease cycle.

However, challenges remain in fully harnessing the potential of DL-based computer vision technologies in agriculture. A key obstacle is the scarcity of high-quality training datasets encompassing diverse crops and diseases, limiting model generalizability and transferability across different regions and conditions. Overcoming this challenge will necessitate collaborative efforts among researchers, farmers, and stakeholders to curate large and varied datasets. In conclusion, our proposed DL framework represents a significant advancement in the development of cost-effective and accessible tools for disease identification and management in precision agriculture. By addressing existing challenges and fostering collaboration, we can further enhance the efficacy and scalability of DL technologies in agricultural settings.

Data availability statement

The data is collected by the authors specifically for this research. Hence, we can make it available to researchers who are interested in similar research upon reasonable request. The data that support the findings of this study are available upon reasonable request from the authors.

Conflict of interest

The authors declare no competing financial interests.

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